

An Automatic Following/Forming Framework for Inverter-Based Resources

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Grid-following (GFL)

- PLL-synchronized current control
- Requires an external voltage-angle reference
- Effective while a strong grid reference exists

Grid-forming (GFM)

- Establishes voltage angle and frequency
- Supports weak-grid and islanded operation
- Can replace the lost reference after SG disconnection

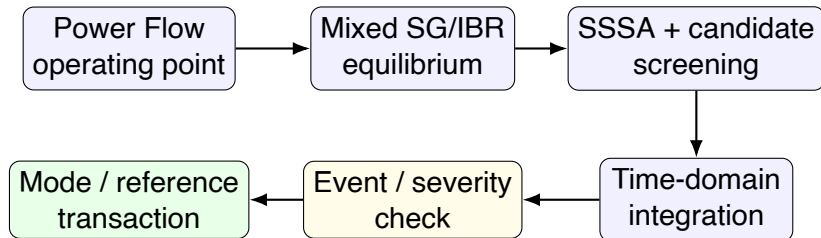
Project Objectives

- 1 Develop one in-house analysis chain for **Power Flow**, **SSSA**, and **Time-Domain Simulation**.
- 2 Model four converter resources with a common plant and selectable **GFL/GFM** controller branches.
- 3 Determine admissible GFM configurations from the network condition instead of fixing the number of forming units in advance.
- 4 Verify automatic mode transitions, island support, and SG reconnection under one disturbance chronology.

Scope

IEEE 14-bus test system, one synchronous generator, four dual-mode IBRs, automatic reference support, and event-driven nonlinear simulation.

Overall Analysis and Switching Workflow



Runtime decision

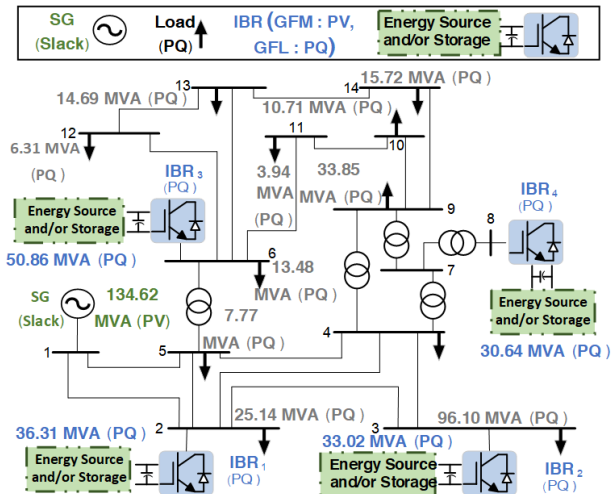
No trigger: advance to the next TS step. Triggered event: certify and commit the mode/reference transaction only after current continuity and right-limit KCL convergence.

Case Study: IEEE 14-Bus with 1 SG + 4 Dual-Mode IBRs

Resource mapping

Resource	Bus
SG1	1
IBR1	2
IBR2	3
IBR3	6
IBR4	8

- SG1: synchronous reference
- IBR1–IBR4: dual-mode converters



Power Flow Formulation

Following the polar form used in Hadi Saadat,

$$Y_{ik} = |Y_{ik}| \angle \theta_{ik}, \quad V_i = |V_i| \angle \delta_i$$

$$P_i = \sum_{k=1}^n |V_i| |V_k| |Y_{ik}| \cos(\theta_{ik} + \delta_k - \delta_i),$$

$$Q_i = - \sum_{k=1}^n |V_i| |V_k| |Y_{ik}| \sin(\theta_{ik} + \delta_k - \delta_i).$$

Bus constraints

Type	Specified	Solved
Slack	$ V , \delta$	P, Q
PV	$P, V $	Q, δ
PQ	P, Q	$ V , \delta$

Newton–Raphson correction

$$\Delta P_i = P_i^{sch} - P_i, \quad \Delta Q_i = Q_i^{sch} - Q_i$$

$$\begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = J_{PF} \begin{bmatrix} \Delta \delta \\ \Delta |V| \end{bmatrix}$$

$$\delta^{(r+1)} = \delta^{(r)} + \Delta \delta^{(r)}$$

$$|V|^{(r+1)} = |V|^{(r)} + \Delta |V|^{(r)}$$

Small-Signal Stability Analysis (SSSA)

Linear small-signal model

$$\Delta \dot{x} = A\Delta x + B\Delta u$$

For the autonomous stability study used here, no AVR/PSS perturbation input is applied:

$$\Delta \dot{x} = A\Delta x$$

The state matrix is obtained from the linearized DAE:

$$A = f_x - f_y (g_y \backslash g_x)$$

Modal analysis

$$A\phi_i = \lambda_i\phi_i$$

$$\lambda_i = \sigma_i + j\omega_i, \quad f_i = \frac{|\omega_i|}{2\pi}$$

$$\zeta_i = -\frac{\sigma_i}{\sqrt{\sigma_i^2 + \omega_i^2}}$$

Stability criterion

$\text{Re}(\lambda_i) < 0$ for every physical mode, together with the declared damping-ratio floor.

$$\dot{x} = f(x, y), \quad 0 = g(x, y)$$

Implicit trapezoidal step

$$R_x = x_{n+1} - x_n - \frac{h}{2} (f_n + f_{n+1}) = 0,$$

$$R_y = g(x_{n+1}, y_{n+1}) = 0,$$

$$J_R \Delta z = -R, \quad z = \begin{bmatrix} x \\ y \end{bmatrix}.$$

Event handling

- Land exactly on each event time.
- Preserve the left-limit state.
- Apply mode/topology changes atomically.
- Require right-limit KCL convergence.
- Roll back a rejected transaction.

Dual-Mode IBR Dynamic Model

Fixed 17-coordinate switching superset

$$\mathbf{x}_{IBR} = [\mathbf{x}_p^T \quad \mathbf{x}_{GFL}^T \quad \mathbf{x}_{GFM}^T \quad I_{dc}]^T$$

$$\mathbf{x}_p = [i_d \ i_q \ V_{dc}]^T$$

$$\mathbf{x}_{GFL} = [\delta_{PLL} \ \xi_{PLL} \ \xi_P \ \xi_Q \ \xi_{Id} \ \xi_{Iq}]^T$$

$$\mathbf{x}_{GFM} = [\delta_{VSG} \ \omega_{VSG} \ E \ \xi_{Vd} \ \xi_{Vq} \ \xi_{Id}^{GFM} \ \xi_{Iq}^{GFM}]^T$$

$$\mathcal{A}_{GFL} = \{1:9, 17\}, \quad \mathcal{A}_{GFM} = \{1:3, 10:17\}.$$

Inactive controller coordinates are held during each mode.

GFL: PLL-synchronized

$$\dot{\delta}_{PLL} = k_{p,PLL} v_q + k_{i,PLL} \xi_{PLL},$$

$$e_P = \kappa P^* - P,$$

$$\dot{i}_d^* = k_{pP} e_P + k_{iP} \xi_P.$$

GFM: voltage-forming VSG

$$M\dot{\omega} = \kappa P^* - P - D_v(\omega - 1),$$

$$\dot{\delta}_{VSG} = \omega_b(\omega - 1),$$

$$\begin{aligned} \tau_E \dot{E} &= k_Q(\kappa Q^* - Q) \\ &\quad - k_E(E - E^*). \end{aligned}$$

Automatic GFL/GFM Switching Law

Severity index

$$J_V = \frac{||V_i| - V_i^*|}{0.10 \text{ p.u.}},$$

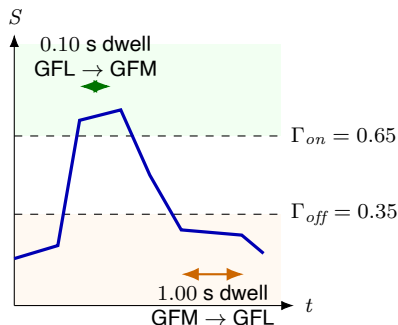
$$J_f = \frac{|f_{COI} - 60|}{0.50 \text{ Hz}},$$

$$S = \text{sat}_{[0,1]}(0.5J_V + 0.5J_f).$$

Hysteretic command

$$q^+ = \begin{cases} \text{GFM}, & S \geq 0.65 \text{ for } 0.10 \text{ s}, \\ \text{GFL}, & S < 0.35 \text{ for } 1.00 \text{ s}, \\ q, & \text{otherwise.} \end{cases}$$

$$||I^+ - I^-||_{\infty} \leq 10^{-10} \text{ p.u.}$$



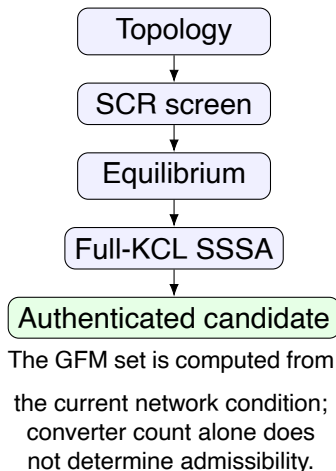
Transition is committed only after equilibrium/SSSA certification and a converged right-limit KCL solve.

Result 1: GFM Candidate Screening

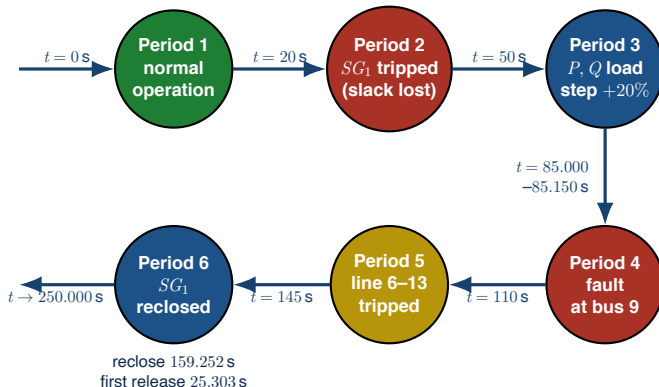
GFM count	Candidate	Outcome
1	any single IBR	admissible
2	IBR1 + IBR3	admissible
2	IBR2 + IBR3	best margin
2	other pairs	rejected
3	any triple	rejected
4	all four IBRs	admissible

Observed property

Admissibility is **non-monotonic** in the number of grid-forming converters.

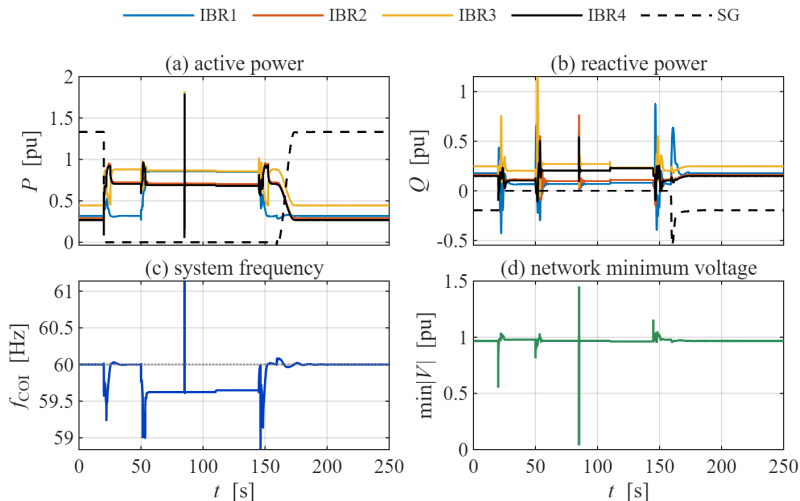


Result 2: Disturbance Scenario



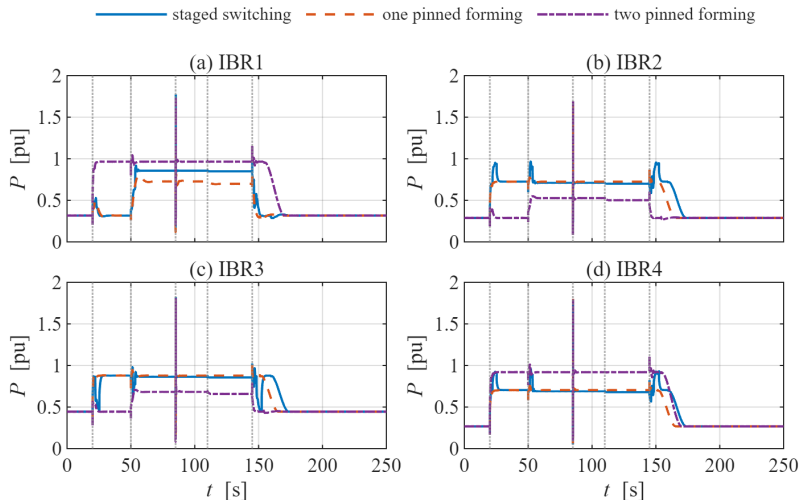
Green: undisturbed; red: source loss or fault; yellow: topology change; blue: restorative action. Scheduled times are case-defined; reclose and release times are read from the accepted event log.

Result 3: Electrical Response



System active/reactive power, COI frequency, and minimum bus voltage over the complete 250 s chronology.

Result 4: Converter Response under Different Policies



IBR1–IBR4 active-power response: staged switching (solid), one pinned GFM (dashed), and two pinned GFMs (dash-dot).

Conclusion and Key Findings

- PF, SSSA, and TS are linked through one nonlinear SG/IBR network model.
- After SG1 is disconnected, at least one voltage-forming IBR is required for the islanded trajectory to continue.
- GFM admissibility is configuration-dependent and non-monotonic in converter count.
- Staged switching reaches the full 250 s chronology and returns all IBRs to GFL operation after SG recovery.

Scope of claim: the staged policy is not claimed to minimize every transient peak; its benefit here is adaptive reference support across the complete disturbance chronology.

Comparative outcome

- **Locked GFL:** stops at 20 s; no forming source.
- **1–2 pinned GFMs:** reach 250 s; reclose succeeds.
- **4 pinned GFMs:** nonlinear failure at 25.485 s.
- **Staged switching:** 250 s with full hand-back.